

A uniqueness theorem for the container derivative

∂ is the only nontrivial representable tangent structure
on finite-support polynomial functors

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Abstract

A companion note showed that the Abbott–Altenkirch–Ghani–McBride one-hole derivative ∂ is the differential combinator of a Cartesian differential category of polynomial functors, realised by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$; that settles *existence*. This note settles *uniqueness*: among representable first-order tangent structures $T = (-) \triangleleft D$ on finite-support polynomial functors there are *exactly two*, the trivial structure $T = \text{id}$ and the container derivative ∂ . The proof has two halves. The first is a rank rigidity: an infinitesimal object $D = y \oplus M$ has a *solid* infinitesimal part when the multiplication $M \triangleleft M \rightarrow M$ is invertible, and over the rig $\mathbb{N}$ the linear containers are automatically free, so solidity reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1. The second is the converse: every representable first-order tangent structure has a solid infinitesimal part. We prove this rig-internally from the universality of the vertical lift, replacing Lanfranchi–Lemay’s ring-theoretic short-exact-sequence step — which needs subtraction — by the observation that in the extensive category of polynomial functors the vertical bundle is a *direct coproduct*, so ranks *add*. The port is thus not only faithful but cleaner: no principal-ideal hypothesis and no subtraction. The finiteness is sharp: $M = \mathbb{N} \cdot y$ is a third solid part over all of Poly , since $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$. The uniqueness answers, and sharpens, a question of Cockett (2012).

1 Introduction

From existence to uniqueness. In a companion note [11] we showed that the category SPoly of finitary polynomial functors is a Cartesian differential category whose differential combinator is the Abbott–Altenkirch–Ghani–McBride one-hole-context derivative ∂ [1], and that the attached tangent bundle is computed by the oldest trick in differential algebra: substitute a dual number and read off the ε -coefficient,

$$T(p) = p(y + \varepsilon v) \bmod \varepsilon^2 = p(y) + \varepsilon v \cdot \partial p.$$

That note answers an *existence* question — there is a tangent structure on polynomial functors whose vertical part is ∂ . The natural next question is *uniqueness*. The dual-number structure is one tangent structure; the identity functor $T = \text{id}$ is another (the trivial one). Are there others? Could a polynomial functor carry an exotic first-order tangent bundle that is neither the derivative nor trivial? This note proves there cannot.

Theorem 1 (Uniqueness). *Among representable first-order tangent structures $T = (-) \triangleleft D$ on finite-support polynomial functors there are exactly two: the trivial structure ($D = y$, $T = \text{id}$) and the AAGM dual-number structure ($D = y + \varepsilon v$, $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$, combinator ∂).*

The question is Cockett’s. That polynomial functors should carry a differential structure is not a new hope. In an unpublished talk at FMCS 2012, *Can you Differentiate a Polynomial?*, Cockett [6] conjectured exactly this — that polynomial functors form a Cartesian differential, hence a tangent, category — fourteen years before the companion note proved it. But the 2012 slides are candid about how much was conjecture: the differentiation there is defined only on “separable” polynomials, closure under composition is asserted but not proved, and the chain rule is flagged as the hard open point (“*of course, need to prove this is a differential!! (chain rule is already challenging...)*”; “*Is this all completely sorted out? Absolutely not!*”). No uniqueness statement appears, nor could it: without the structure in hand there is nothing to count. The companion note supplied the structure; the present note supplies the count, and so closes the arc Cockett opened. (We flag one divergence in §9: combinatorial species, the setting nearest Cockett’s, are most naturally a tangent *bicategory* [10], structurally different from the SPoly one-category we work in.)

The template, and why it does not simply transfer. For commutative algebra the uniqueness question has a sharp answer. Lanfranchi and Lemay [9] classify the *representable* tangent structures on the opposite of the category of modules (equivalently, affine schemes): a representable tangent structure is a tensoring $T = D \otimes (-)$ by an *infinitesimal object* $D = R \oplus M$, and when the base ring R is a principal ideal domain there are *exactly two* — the trivial one ($M = 0$) and the Kähler/dual-number one ($M = R$). Their argument has two load-bearing uses of ring structure. The *rigidity* half (a solid infinitesimal part has rank n with $R^{n^2} \cong R^n$, forcing $n^2 = n$, hence $n \in \{0, 1\}$; their Theorem 4.17, Corollary 4.18) needs R to be a PID, so that the infinitesimal module is free with a well-defined rank. The *converse* half (a tangent structure’s infinitesimal part *is* solid; their Theorem 4.8, via the universality of the vertical lift) computes a vertical bundle by a short exact sequence — i.e. by *subtraction* of ranks. Neither transfers to polynomial functors verbatim: the base there is the rig $\mathbb{N}$, which is not a PID and has no subtraction.

What we prove, and the one new idea. We prove Theorem 1 by porting *both* halves to the substitution monoidal category $(\text{Poly}, \triangleleft, y)$, and in both the port turns out *cleaner* than its template.

- *Rigidity* (§4). An infinitesimal part is a *linear* container $M = S \cdot y$, forced by the additive-bundle axiom. Over $\mathbb{N}$ a linear container simply *is* a free module — the linear polynomial functors are by definition the free $\mathbb{N}$ -modules — so freeness is *automatic*, with no PID hypothesis. Solidity then reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1: the finite solid parts are exactly $\{0, y\}$.
- *Converse* (§6). Every representable first-order tangent structure has a solid infinitesimal part. Here is the one new idea. The place Lanfranchi–Lemay use subtraction is the computation of the vertical bundle fibre P' by exactness. In the *extensive* category of polynomial functors that fibre is instead a *direct coproduct* of the surviving summands,

$$P' = y \oplus \varepsilon_{\text{out}} \oplus (M \triangleleft M),$$

so its rank is a *sum*, $1 + n + n^2$, computed by addition and never as a difference. The universality of the vertical lift (Cockett–Cruttwell [7]) identifies the second tangent bundle T_2 with the vertical bundle V ; evaluating this functor isomorphism at y gives $D_2 \cong P'$, i.e. $1 + 2n = 1 + n + n^2$, and the cancellative rig $\mathbb{N}$ (no zero divisors, *no subtraction*) forces $n^2 = n$, hence $n \in \{0, 1\}$.

Assembling the two halves with the forward correspondence (both surviving parts are realised — id and ∂) and representability [11] gives Theorem 1.

The finiteness is sharp. Unqualified uniqueness over all of Poly is false, and we exhibit the obstruction explicitly: $M = \mathbb{N} \cdot y$ is solid, because $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y$ by the Cantor pairing (§5, Proposition 9); and the converse’s rank equation $1 + 2\kappa = 1 + \kappa + \kappa^2$ is satisfied by every infinite cardinal κ , so universality does not exclude it either. This is the precise analogue of Lanfranchi–Lemay’s extra structures when finite generation is dropped (their Examples 4.20–4.21), and it tells us the load-bearing hypothesis is a *finite-dimensional* infinitesimal part — finitely many tangent directions — here taken as plain finiteness of the shape set.

Certification. The arithmetic of both halves is machine-checked in a Mathlib-free Lean 4 development with no classical axioms: the idempotent collapse $n^2 = n \iff n \in \{0, 1\}$, the finite enumeration of solid ranks, the three coproduct-rank identities underwriting the “ranks add, never subtract” claim, and the universality collapse $1 + 2n = 1 + n + n^2 \iff n \in \{0, 1\}$ by additive cancellation. The Lean certifies the *arithmetic*; the structural inputs (the universality axiom, that $\triangleleft$ preserves pullbacks, that M is linear) are cited framework facts, and we are explicit about the boundary (§4, §6).

Contribution. The contributions are: the uniqueness theorem itself (Theorem 1), proved over a rig with neither a PID hypothesis nor subtraction; the rigidity half (finite solid parts = $\{0, y\}$); the rig-internal converse, whose single new idea is that extensivity turns the vertical bundle into a coproduct so that ranks add (§6); the technical observation that on linear containers the *bundle* tensor $\triangleleft$ and the *module* (Dirichlet) tensor coincide (Lemma 4), so solidity is computed in the bundle’s own tensor; the sharpness of the finiteness hypothesis (Proposition 9); and the Lean certification of the arithmetic. The result is a clean deliverable for the tangent-category and Poly communities: *the container derivative is the only nontrivial representable first-order tangent structure a finite-support polynomial functor can carry*.

Outline. §2 recalls containers, the substitution monoidal structure, infinitesimal objects, the representable tangent bundle and the universality of the vertical lift. §3 proves the bundle/module tensor coincidence. §4 proves the rank rigidity (finite solid parts = $\{0, y\}$) and records its Lean certification. §5 proves the infinite-support obstruction. §6 proves the solidity converse from vertical-lift universality — the main new content — and records its Lean certification. §7 assembles the uniqueness theorem. §8 compares with Lanfranchi–Lemay. §9 concludes with the precedent and the sequel.

2 Preliminaries

We recall only what the argument uses; for the Cartesian differential and tangent structure on SPoly we refer to the companion note [11].

Containers and the substitution monoidal category. A *container in n variables* is a set S of *shapes* together with, for each $s \in S$ and each $i \in \{1, \dots, n\}$, a position set P_s^i ; its extension is the polynomial functor $\llbracket S \triangleleft P \rrbracket(\vec{X}) = \sum_{s \in S} \prod_{i=1}^n X_i^{P_s^i}$. We work with finitary containers, so only the cardinalities $|P_s^i|$ matter up to isomorphism [1]. Composition of containers is substitution

of extensions, written $G \triangleleft F$, with unit the identity container y ; this makes $(\mathbf{Poly}, \triangleleft, y)$ a (strict, once we pass to isomorphism classes) monoidal category, and the comonoids in it are exactly the small categories [2, 8]. A container is *linear* if it is of the form $S \cdot y$ for a set S , i.e. its extension is $\vec{X} \mapsto S \times X$ in one variable; *finite-support* if its shape set is finite. We write $\mathbf{SPoly}$ for the finite-support polynomial functors.

Infinitesimal objects and representable tangent bundles. We use the representable presentation of tangent structure of [9], transported along the dictionary $(\mathbf{Mod}_R, \otimes_R, R) \rightsquigarrow (\mathbf{Poly}, \triangleleft, y)$.

Definition 2 (Infinitesimal object). An *infinitesimal object* in $(\mathbf{Poly}, \triangleleft, y)$ is a polynomial D equipped with an augmentation: morphisms $z: y \rightarrow D$ (zero) and $p: D \rightarrow y$ (projection) with $p \circ z = \text{id}_y$. The splitting $D = y \oplus M$ defines the *infinitesimal part* $M = \ker(p)$. Right- $\triangleleft$ -tensoring by D is the endofunctor $T := (-) \triangleleft D$, with tangent projection $A \triangleleft p: A \triangleleft D \rightarrow A$ and zero section $A \triangleleft z: A \rightarrow A \triangleleft D$.

Definition 3 (Representable first-order tangent structure). A *representable first-order tangent structure* is a tangent structure (in the sense of Cockett–Cruttwell [7], first order) whose bundle functor is $T = (-) \triangleleft D$ for an infinitesimal object $D = y \oplus M$. Its infinitesimal part M is called *solid* if the tangentoid multiplication $\mu: M \triangleleft M \rightarrow M$ is an isomorphism.

The dual-number point $D = y + \varepsilon v$ (read as the infinitesimal object of the base change $\mathbb{N} \rightarrow \mathbb{N}[\varepsilon]/\varepsilon^2$) is such an object: the augmentation is $\varepsilon \mapsto 0$, the zero is $y \hookrightarrow D$, and $(-) \triangleleft D$ is exactly the dual-number substitution T of [11]. Its infinitesimal part is the linear container $M = \langle \varepsilon v \rangle = 1 \cdot y$, of rank 1; the trivial structure is $D = y$, $M = 0$, $T = \text{id}$. That $T = (-) \triangleleft D$ is a genuine tangent structure — not merely an endofunctor — is the content of [11]; we take representability of ∂ as established there and ask only whether other infinitesimal objects give rise to further structures.

The universality of the vertical lift. Among the tangent-structure axioms of [7] we use one by name. The *second tangent bundle* is $T_2 := T \times_{\text{id}} T$ (the pullback of $T \xrightarrow{p} \text{id}$ along itself), and the *vertical bundle* V is the pullback of the tangent map $Tp: T^2 \rightarrow T$ along the zero $0: \text{id} \rightarrow T$. The *vertical lift* is a canonical comparison $T_2 \rightarrow T^2$, and the axiom of *universality of the vertical lift* ([7], part of the definition of a tangent structure) asserts that the square

$$\begin{array}{ccc} T_2 & \longrightarrow & T^2 \\ \downarrow & & \downarrow Tp \\ \text{id} & \xrightarrow{0} & T \end{array}$$

is a pullback; equivalently, that the induced map $T_2 \rightarrow V$ is an isomorphism. This is the single structural axiom the converse of §6 invokes.

Why the infinitesimal part is linear. For a tangent structure the projection $p: T \Rightarrow \text{id}$ is an *additive bundle*: each fibre carries a commutative-monoid structure and the bundle addition is linear. For $T = (-) \triangleleft D$ this forces the infinitesimal part M to be a *linear* container $M = S \cdot y$. A non-linear M — for instance εy^2 , whose ε -coefficient is $y^2 p'$ rather than a directional $v \cdot p'$ — has no vector-space fibre and fails the additive-bundle axiom. So linearity of M is *forced*, not assumed, and it is this linearity (not any projective-module theory) that supplies freeness in §4.

3 The bundle and module tensors coincide on the linear layer

The port of the Lanfranchi–Lemay argument has one notorious hazard: the tangent *bundle* is built from substitution $\triangleleft$, whereas the *solid* condition $M \otimes M \cong M$ lives in the “module layer” where rank multiplies, and these look like two different monoidal products. One might fear a mismatch between the tensor the bundle uses and the tensor in which solidity is measured. The following observation dissolves the fear.

Lemma 4 (Tensor coincidence on the linear layer). *For linear containers, substitution and the Dirichlet product agree:*

$$(S \cdot y) \triangleleft (T \cdot y) = (S \times T) \cdot y = (S \cdot y) \otimes_{\text{Dir}} (T \cdot y).$$

Proof. The extension of $S \cdot y$ is $X \mapsto S \times X$. Composing, $(S \cdot y) \triangleleft (T \cdot y): X \mapsto S \times (T \times X) = (S \times T) \times X = (S \times T) \cdot y$. The Dirichlet product of $\sum_s y^1$ and $\sum_t y^1$ is $\sum_{(s,t)} y^{1 \times 1} = (S \times T) \cdot y$. Both equal $(S \times T) \cdot y$. $\square$

Remark 5. The two tensors agree on monomials ($y^a \triangleleft y^b = y^{ab} = y^a \otimes_{\text{Dir}} y^b$), and in particular on the linear layer, which is all the rigidity needs: the infinitesimal part is linear (§2), and $M \triangleleft M$ is computed there. On polynomials with several monomials the two diverge — already $(y + 1) \triangleleft (y + 1) = y + 2$ while $(y + 1) \otimes_{\text{Dir}} (y + 1) = y + 3$ — so Lemma 4 is genuinely a statement about the linear layer, not a global identity. Neither tensor is the “doubling” Dirichlet action $(2y) \otimes_{\text{Dir}} (-)$, which sends $p \mapsto 2p$ and has nothing to do with the derivative.

Consequence. On the linear layer the *bundle* tensor $\triangleleft$ already multiplies rank, so the solid condition $M \triangleleft M \cong M$ reads $|S|^2 = |S|$ in the very tensor the bundle uses. There is no mismatch; this single coincidence is what makes the port faithful.

4 Rank rigidity: the solid parts are $\{0, y\}$

We classify the finite solid infinitesimal parts. Two structural facts about linear containers, then the rank computation.

Lemma 6 (Freeness is automatic over $\mathbb{N}$). *Every linear container $S \cdot y$ is the free $\mathbb{N}$ -module on the set S ; two linear containers are isomorphic iff their shape sets are equinumerous. The cardinal $|S| =: \text{rank}(M)$ is a complete isomorphism invariant.*

Proof. Under coproduct of containers and the $\mathbb{N}$ -action $n \cdot (S \cdot y) = (n \times S) \cdot y$, the linear containers form the free- $\mathbb{N}$ -module functor applied to sets: $S \cdot y$ is the $|S|$ -fold copower of y . A container isomorphism $S \cdot y \cong T \cdot y$ is, since morphisms between linear containers are just functions on shapes (positions being singletons), a bijection $S \cong T$. Hence rank is a complete invariant. $\square$

This is the analogue of Lanfranchi–Lemay’s “projective $\Rightarrow$ free $\Rightarrow$ rank is a complete invariant” (their Lemma 4.16, via Rotman’s structure theorem, which *requires* the base to be a PID). Here it is *automatic*: the linear polynomial functors *are* the free $\mathbb{N}$ -modules by definition, so no PID hypothesis is needed and the rigidity below holds over the rig $\mathbb{N}$ — strictly outside the Lanfranchi–Lemay hypotheses.

Lemma 7 (Solid $\Rightarrow$ idempotent rank). *If $M = S \cdot y$ is solid, i.e. $\mu: M \triangleleft M \rightarrow M$ is an isomorphism, then $|S|^2 = |S|$.*

Proof. By Lemma 4, $M \triangleleft M = (S \times S) \cdot y$, of rank $|S|^2$ by Lemma 6. An isomorphism $\mu: (S \times S) \cdot y \cong S \cdot y$ forces, by Lemma 6, a bijection $S \times S \cong S$, i.e. $|S|^2 = |S|$. $\square$

Theorem 8 (Finite rigidity). *A finite-support solid infinitesimal part has rank in $\{0, 1\}$. Hence there are exactly two: $M = 0$ (rank 0) and $M = y$ (rank 1). By §2 these are the trivial structure and the AAGM dual-number structure, with combinator ∂ .*

Proof. For a natural number $n = |S|$ we have $n^2 = n \iff n(n-1) = 0 \iff n \in \{0, 1\}$, since $\mathbb{N}$ has no zero divisors. By Lemma 7 a finite solid part has $|S|^2 = |S|$, so $|S| \in \{0, 1\}$; by Lemma 6 the rank determines M , giving $M \in \{0, y\}$. $\square$

This is the Lanfranchi–Lemay rank computation $R^{n^2} \cong R^n \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$, ported verbatim with the cardinality rank, and — per Lemma 6 — without their PID hypothesis.

Machine certification of the arithmetic. The finite arithmetic of Theorem 8 is certified in a Mathlib-free Lean 4 development with no classical axioms. The oracle proves: $n^2 = n \iff n \in \{0, 1\}$ on $\mathbb{N}$ (in general, `omega-free`); that the solid ranks in $0 \leq n \leq 12$ are exactly $\{0, 1\}$ (by `decide`); that these two ranks map through the structure assignment to exactly $\{\text{id}, \partial\}$, with the dual-number object of arity 2; that every rank $r \geq 2$ (the Whitney sums of §6) is non-solid; and that the Cantor pairing is injective on a 12×12 window, witnessing $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ for §5. `#print axioms` on the packaged oracle reports only `propext` (a structural, non-classical axiom); the finite `decide` claims are individually axiom-free, with no `sorryAx`, `Quot.sound` or `Classical.choice`.

5 The finiteness hypothesis is sharp

Theorem 8 is stated for finite support, and the hypothesis cannot be dropped.

Proposition 9 (Infinite-support obstruction). *The countable linear container $M = \mathbb{N} \cdot y$ is solid: $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y = M$, because $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$. It is a solid infinitesimal part distinct from 0 and y . Consequently unqualified uniqueness over all of `Poly` is false.*

Proof. Lemma 4 gives $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y$, and the Cantor pairing supplies a bijection $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$ (injectivity checked on a 40×40 window; see the Lean oracle of §4 for a certified 12×12 window). By Lemma 6 this is an isomorphism $\mu: M \triangleleft M \cong M$, and $\text{rank } M = |\mathbb{N}| \notin \{0, 1\}$. $\square$

More generally every infinite cardinal κ satisfies $\kappa^2 = \kappa$ (under choice), so each yields a solid infinitesimal part. This is the exact analogue of Lanfranchi–Lemay’s extra structures when the finite-generation hypothesis is relaxed — their non-PID example $R = S \times S$ (Example 4.21) and non-finitely-generated $\mathbb{Z} \times \mathbb{Q}$ (Example 4.20). The moral is that the honest hypothesis is a *finite-dimensional* infinitesimal part: finitely many tangent directions, matching Lanfranchi–Lemay’s finite-generation (Niefield coexponentiability) condition, here realised as plain finiteness of the shape set. We will see in §6 that the converse does not exclude $\mathbb{N} \cdot y$ either, so finiteness is load-bearing for *both* halves of the uniqueness theorem.

6 The solidity converse: universality of the vertical lift

Theorem 8 classifies the solid infinitesimal *parts*. To pass from parts to tangent *structures* we need the converse direction of the correspondence

($\star$) representable first-order tangent structure $T = (-) \triangleleft D \iff$ solid infinitesimal part $M = \ker(D \rightarrow y)$.

The forward direction is immediate once the parts are known (§7). The converse — that every representable first-order tangent structure has a *solid* infinitesimal part — is Lanfranchi–Lemay’s Theorem 4.8, proved over a commutative ring, and is the step that did not transfer verbatim: their derivation computes a vertical bundle by a short exact sequence, hence by subtraction, which the rig $\mathbb{N}$ lacks. We now give a rig-internal proof. The engine is the universality of the vertical lift (§2); the one new idea is that, in the extensive category of polynomial functors, the vertical bundle is a *direct coproduct*, so its rank is computed by addition and never by a difference.

Throughout, $T = (-) \triangleleft D$ with $D = y \oplus M$, $M = S \cdot y$ linear of finite rank $n = |S|$ (linearity forced, §2).

6.1 Substitution preserves pullbacks on both sides

Lemma 10 ($\triangleleft$ preserves pullbacks). *For every $D \in \mathbf{Poly}$ the functor $(-) \triangleleft D$ preserves pullbacks; and for every polynomial A the functor $A \triangleleft (-)$ preserves pullbacks.*

Proof. Pullbacks in $\mathbf{Poly}$ are computed pointwise. For $(-) \triangleleft D$: if $F = G \times_H K$ pointwise then for all X , $(F \triangleleft D)(X) = F(D(X)) = G(D(X)) \times_{H(D(X))} K(D(X)) = (G \triangleleft D \times_{H \triangleleft D} K \triangleleft D)(X)$, so $F \triangleleft D = G \triangleleft D \times_{H \triangleleft D} K \triangleleft D$. For $A \triangleleft (-)$: a polynomial functor A preserves connected limits ([8], Prop. 1.9), and a pullback is a connected limit, so A preserves the pointwise pullback $E_1(X) \times_{E_0(X)} E_2(X)$; hence $A \triangleleft (-)$ preserves pullbacks. $\square$

Consequence: the tangent data are all right- $\triangleleft$ -tensorings. The projection, zero, tangent map and the two bundles of $T = (-) \triangleleft D$ are $(-) \triangleleft$ (structure map on D): $p = (-) \triangleleft \hat{p}$ with $\hat{p}: D \rightarrow y$ the augmentation and $z = (-) \triangleleft \hat{z}$ with $\hat{z}: y \rightarrow D$ the zero; and, by Lemma 10 applied so that $A \triangleleft (-)$ carries the relevant pointwise pullback,

$$T_2 = (-) \triangleleft D_2, \quad D_2 := D \times_y D; \quad V = (-) \triangleleft P', \quad P' := \hat{g}^{-1}(\mathrm{im} \hat{z}), \quad \hat{g} := \hat{p} \triangleleft D: D \triangleleft D \rightarrow D.$$

All of D , D_2 , $D \triangleleft D$, P' are objects of $\mathbf{Poly}$, and y is one of them.

6.2 The three coproduct counts

We work in the based setting: ε -grading modulo ε^2 within each copy of D , with the mixed term surviving across copies — the Weil-algebra model $\mathbb{N}[\varepsilon_1, \dots]/(\text{within-copy squares})$ that the base-change presentation of T [11] delivers. In the extensive category $\mathbf{Poly}$ each object below is a direct coproduct of summands, so its rank is the *sum* of the summand ranks.

Lemma 11 (Coproduct ranks). *With $n = \mathrm{rank} M$ and $\mathrm{rank}(M \triangleleft M) = n^2$ (Lemmas 4, 6):*

$$\begin{aligned} D \triangleleft D &= y \oplus M \oplus M \oplus (M \triangleleft M), & \mathrm{rank} &= 1 + n + n + n^2 = (1 + n)^2; \\ D_2 &= D \times_y D = y \oplus M \oplus M, & \mathrm{rank} &= 1 + 2n; \\ P' &= y \oplus \varepsilon_{\mathrm{out}} \oplus (M \triangleleft M), & \mathrm{rank} &= 1 + n + n^2. \end{aligned}$$

Proof. ($D \triangleleft D$.) Since M is linear, $M(X) = S \times X$, so $(D \triangleleft D)(X) = D(D(X)) = X \oplus M(X) \oplus M(X) \oplus M(M(X))$: the base y , the inner copy’s M ($\varepsilon_{\mathrm{in}}$), the outer copy’s M ($\varepsilon_{\mathrm{out}}$), and the mixed $M \triangleleft M = (S \times S) \cdot y$. Ranks $1, n, n, n^2$.

(D_2 .) A fibre product over the base shares y and adjoins the two infinitesimal directions but creates *no* product of the two infinitesimals: $D_2 = y \oplus M \oplus M$, rank $1 + 2n$.

(P' .) The map $\hat{g} = \hat{p} \triangleleft D$ applies the augmentation $\hat{p} (\varepsilon \mapsto 0)$ to the *outer* copy: it kills ε_{out} and the mixed term $M \triangleleft M$ (both carry an outer- M factor) and fixes y and ε_{in} . The zero $\hat{z}$ has image the y -summand. Hence $P' = \hat{g}^{-1}(\text{im } \hat{z})$ is the sub-coproduct on which the ε_{in} -component vanishes: $P' = y \oplus \varepsilon_{\text{out}} \oplus (M \triangleleft M)$, rank $1 + n + n^2$. In the extensive category **Poly** this pullback *is* the sub-coproduct of surviving summands, so its rank is the sum $1 + n + n^2$ — not $\text{rank}(D \triangleleft D) - \text{rank}(\text{im } \hat{z})$. This is the step where Lanfranchi–Lemay form a difference of ranks (a short exact sequence over a ring), and where we do not. $\square$

Both $D_2 = (1 + 2n) \cdot y$ and $P' = (1 + n + n^2) \cdot y$ are linear (a coproduct $y \oplus (\text{free})$ is again linear), so Lemma 6 applies to them.

6.3 Universality forces $n^2 = n$

Theorem 12 (Solidity converse). *Let $T = (-) \triangleleft D$ be a representable first-order tangent structure on finite-support polynomial functors, with $D = y \oplus M$ and $n = \text{rank } M$ finite. Then $n^2 = n$, hence $n \in \{0, 1\}$; i.e. M is solid and $M \in \{0, y\}$.*

Proof. By the universality of the vertical lift (§2), the comparison $T_2 \rightarrow V$ is an isomorphism of endofunctors, naturally in the argument. By Lemma 10 and its consequence, $T_2 = (-) \triangleleft D_2$ and $V = (-) \triangleleft P'$. Evaluate the functor isomorphism at the object $y \in \mathbf{Poly}$ and use $y \triangleleft E = E$:

$$D_2 \cong P' \quad \text{in } \mathbf{Poly}.$$

Both are linear (Lemma 11), so by Lemma 6 their shape-counts agree:

$$1 + 2n = 1 + n + n^2.$$

Now use only that $\mathbb{N}$ is a cancellative rig with no zero divisors — *no subtraction, no additive inverses*. Additive left-cancellation of the common summand $1 + n$ gives $n = n^2$. For $n \geq 1$, multiplicative cancellation of n in $n \cdot n = n \cdot 1$ gives $n = 1$; and $n = 0$ satisfies $0^2 = 0$. Hence $n \in \{0, 1\}$, so $M \in \{0, y\}$ by Lemma 6, and $\mu: M \triangleleft M \rightarrow M$ is an isomorphism in each case. $\square$

Remark 13 (Why this is rig-internal). The subtraction that blocked a literal port lived in computing the vertical-bundle fibre P' by exactness. In **Poly** that fibre is a direct coproduct of the surviving summands (Lemma 11), so its rank is a *sum*. Everything downstream is additive/multiplicative cancellation in $\mathbb{N}$. The derivation never forms a difference of ranks, never inverts an element, and never constructs the solid map μ explicitly — solidity is *output*, read off from $n \in \{0, 1\}$. Lanfranchi–Lemay’s Theorem 4.8 instead builds μ and argues over a ring; the content we have replaced is precisely their one subtraction.

Remark 14 (The non-solid finite candidates are auxiliary bundles). The finite candidate objects excluded by Theorem 12 are the higher square-zero objects $D_r = y + \varepsilon_1 v_1 + \cdots + \varepsilon_r v_r$ ($\varepsilon_i \varepsilon_j = 0$, $r \geq 2$), with $M = \mathbb{N}^r$, $M \triangleleft M = \mathbb{N}^{r^2}$, rank $r^2 \neq r$: not solid. These D_r are precisely the Whitney sums $T_r = T \times_X \cdots \times_X T$, which occur in the tangent axioms as *auxiliary* bundles, not as standalone tangent structures. So the objects the converse rejects are exactly the ones that are not primitive structures — solidity coincides with indecomposability, as it should. (The non-solidity of every $r \geq 2$ is part of the certified Lean oracle below.)

Remark 15 (Consistency with sharpness). The rank equation $1 + 2n = 1 + n + n^2$ is satisfied by every infinite cardinal κ ($1 + 2\kappa = \kappa = 1 + \kappa + \kappa^2$), so universality does *not* exclude $M = \mathbb{N} \cdot y$. The converse is therefore as sharp as the rigidity: both hold only under finiteness, consistent with Proposition 9.

Machine certification of the arithmetic. The rig-internal arithmetic of Theorem 12 is certified in the same Mathlib-free, axiom-light Lean 4 style. The oracle proves the three coproduct-rank identities of Lemma 11 — $\text{rank}(D \triangleleft D) = (1 + n)^2$, $\text{rank}(D \triangleleft D) = \text{rank } P' + n$, $\text{rank}(D \triangleleft D) = \text{rank } D_2 + n^2$ — which are the honesty certificate for “ranks add, never subtract”; the universality collapse $\text{rank } D_2 = \text{rank } P' \iff n \in \{0, 1\}$, proved by `Nat.add_left_cancel` (literal additive cancellation, *no omega*) reducing to $n^2 = n \iff n \in \{0, 1\}$; the finite enumeration (survivors in $0 \leq n \leq 12$ are exactly $\{0, 1\}$); and the two witnesses $n = 1$ (universality holds, $\text{rank } D_2 = \text{rank } P' = 3$, the Weil rank of $\mathbb{N}[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2, \varepsilon_1 \varepsilon_2)$) and $n = 2$ (excluded, $5 \neq 7$). `#print axioms` on the packaged oracle reports only `propext`. The Lean certifies the *arithmetic collapse* and the additivity of the counts; it does *not* formalise the structural inputs — the universality axiom ([7]), that $\triangleleft$ preserves pullbacks (Lemma 10, via [8]), or that M is linear (the additive-bundle axiom) — which are cited framework facts, and we are explicit that they are the mathematical, not the decidable, content.

7 The uniqueness theorem

With both halves in hand the correspondence $(\star)$ is an equivalence, and the uniqueness theorem assembles.

Forward: solid $\Rightarrow$ structure. A finite solid part yields a tangent structure in the two surviving cases. $M = 0$ gives $T = \text{id}$; $M = y$ gives the AAGM dual-number structure, which *is* a tangent structure — the full Cartesian differential / tangent verification is the content of [11]. Since Theorem 8 leaves no third finite solid part, the forward direction never has to manufacture a third structure.

Theorem 16 (Uniqueness). *Among representable first-order tangent structures $T = (-) \triangleleft D$ on finite-support polynomial functors there are exactly two: the trivial structure ($D = y$, $T = \text{id}$) and the AAGM dual-number structure ($D = y + \varepsilon v$, $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$, combinator ∂).*

Proof. Let $T = (-) \triangleleft D$ be such a structure, $D = y \oplus M$. By Theorem 12 its infinitesimal part is solid with $\text{rank } M \in \{0, 1\}$; by Theorem 8 (equivalently Lemma 6) this forces $M \in \{0, y\}$, hence $D \in \{y, y + \varepsilon v\}$ up to isomorphism. By the forward direction both are realised, by id and ∂ respectively. Hence exactly two. $\square$

This is Theorem 1, now unconditional: the conditional hypothesis of the earlier draft — that every representable first-order tangent structure has a solid infinitesimal part — is discharged by Theorem 12.

8 Comparison with Lanfranchi–Lemay

The theorem is a clean deliverable for two audiences. For the tangent-category community [7, 9] it exhibits a rig-based model in which the representable-tangent rigidity *and* its vertical-lift converse hold for strictly weaker reasons than over a PID — no principal ideals, no subtraction — isolating exactly which uses of ring structure in [9] were essential (none, for finite-dimensional infinitesimal parts) and which were incidental (the PID and the short exact sequence, both replaced). For the Poly and containers community [1, 2] it upgrades “the container derivative is a tangent structure” [11] to “it is the *only* nontrivial one,” a uniqueness theorem for a construction that is otherwise defined by a formula.

Lanfranchi–Lemay (affine schemes)	This note (polynomial functors)
base a commutative ring R ; rigidity needs R a PID	base the rig $\mathbb{N}$ (no subtraction); rigidity needs no PID
projective $\Rightarrow$ free = Rotman structure theorem (uses PID)	linear functor <i>is</i> free $\mathbb{N}$ -module (Lemma 6) — automatic
rank in R -modules, $R^{n^2} \cong R^n$	cardinality rank, $(S \times S) \cong S$ (Lemmas 4, 6)
rigidity: solid $M \otimes_R M \cong M \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$	rigidity: solid $M \triangleleft M \cong M \Rightarrow S ^2 = S \Rightarrow S \in \{0, 1\}$
converse: vertical bundle fibre via short exact sequence (rank by subtraction)	converse: fibre P' a direct coproduct $y \oplus \varepsilon_{\text{out}} \oplus M \triangleleft M$ (rank by addition ; Lemma 11)
two structures: trivial + Kähler/dual-number extras at non-PID / non-f.g. (Ex. 4.20–4.21)	two structures: $\text{id} + \partial$ (Theorem 16) extra at infinite support $\mathbb{N} \cdot y$ (§5)

Table 1: The port is faithful (same rank-idempotent mechanism, same vertical-lift converse) and sharper on two axes: freeness is automatic over $\mathbb{N}$ (rigidity over a rig, not a PID), and the vertical bundle is a coproduct (converse by addition, not subtraction). The tensor coincidence (Lemma 4) and the extensivity of **Poly** (Lemma 11) are the linchpins specific to polynomial functors.

9 Conclusion: precedent and sequel

We have proved that the container derivative ∂ is the unique nontrivial representable first-order tangent structure a finite-support polynomial functor can carry (Theorem 16). The proof ports Lanfranchi–Lemay’s affine-scheme classification in both halves, and in both the rig $\mathbb{N}$ makes the port *cleaner*: freeness is automatic, so the rigidity needs no PID (Lemma 6); and the vertical bundle is a direct coproduct, so the converse needs no subtraction (Lemma 11, Theorem 12). Two technical observations carry the port: on the linear layer the bundle tensor $\triangleleft$ and the module tensor coincide (Lemma 4), and **Poly** is extensive so that ranks add. The arithmetic of both halves is Lean-certified with no classical axioms. The finiteness is sharp: $\mathbb{N} \cdot y$ is a third solid part over all of **Poly**, and neither half of the theorem excludes it, so the honest hypothesis is a finite-dimensional infinitesimal part.

Precedent. The hope that polynomial functors carry a differential structure is Cockett’s. His unpublished FMCS 2012 talk *Can you Differentiate a Polynomial?* [6] conjectured that they form a Cartesian differential, hence a tangent, category — the thesis the companion note [11] later proved. The 2012 slides, though, are explicit that this was conjecture: the differentiation there is defined only on “separable” polynomials, closure under composition is asserted rather than proved, and the chain rule is named as the hard open point. (The slides even record, as an aside, that this is “a differential category in which negation is unnatural” — the same absence of additive inverses that we have just turned to advantage, since it is exactly what makes the vertical-bundle coproduct, rather than a short exact sequence, the right tool.) No uniqueness statement appears in 2012. The present note supplies it, and so answers Cockett’s question in its strongest form: not only *can* you differentiate a polynomial, but differentiation is the *only* nontrivial way to do so. We flag one caveat, due to Cruttwell via [10]: combinatorial species — the setting closest to Cockett’s — are most naturally a tangent *bicategory*, with species as 1-morphisms, which is structurally different from the **SPoly** one-category treated here; our uniqueness is a statement about the latter.

Sequel. A rigid tangent structure is the entry condition for the Cartan-calculus / Lie-algebroid program of Aintablian and Blohmann [3], which asks when a tangent category supports a scalar “ring object” R making each TX an R -module compatibly with the tangent axioms, and thereby upgrades a tangent structure to a Lie-algebroid structure with a full Cartan calculus. That $\mathbf{SPoly}$ now has a *unique* nontrivial first-order tangent structure makes the question well-posed: is there such an R for $(\mathbf{SPoly}, \partial)$? The negation remark above is a warning sign — a scalar ring object requires additive inverses, which $\mathbf{N}$ does not have — and determining exactly how much of the Cartan calculus survives over a rig is the natural next problem. We leave it, together with higher Weil algebras $\mathbb{N}[\varepsilon]/\varepsilon^k$ ($k \geq 3$) and iterated tangents (first-order out of scope here, as for [1, 9, 5]), to future work.

References

- [1] M. Abbott, T. Altenkirch, N. Ghani, C. McBride, *Derivatives of containers*, Typed Lambda Calculi and Applications (TLCA 2003), LNCS 2701, Springer, 2003, 16–30.
- [2] D. Ahman, T. Uustalu, *Directed containers as categories*, Proc. MSFP 2016, EPTCS 207, 2016, 89–98.
- [3] A. Aintablian, C. Blohmann, *Cartan calculus in tangent categories*, preprint, arXiv:2607.11169 (2026).
- [4] R. F. Blute, J. R. B. Cockett, R. A. G. Seely, *Cartesian differential categories*, Theory and Applications of Categories **22** (2009), no. 23, 622–672.
- [5] M. Capucci, G. Cruttwell, N. Ghani, F. Zanasi, *A fibrational theory of first order differential structures*, preprint, arXiv:2409.05763 (2024).
- [6] J. R. B. Cockett, *Can you Differentiate a Polynomial?*, slides, Foundational Methods in Computer Science (FMCS 2012), unpublished.
- [7] J. R. B. Cockett, G. S. H. Cruttwell, *Differential structure, tangent structure, and SDG*, Applied Categorical Structures **22** (2014), no. 2, 331–417.
- [8] N. Gambino, J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 1, 153–192.
- [9] M. Lanfranchi, J.-S. P. Lemay, *Representable tangent structures for affine schemes*, preprint, arXiv:2505.09080 (2025).
- [10] *Combinatorial species and differential categories*, MathOverflow question 341933, mathoverflow.net/q/341933 (accepted answer relaying G. Cruttwell; accessed 2026).
- [11] MacBeth, *The container derivative is a tangent structure: a fibrational positioning* (Kodamai internal note, 2026).